

GRAFT VERSUS HOST DISEASE

Lipocalin-2 expression identifies an intestinal regulatory neutrophil population during acute graft-versus-host disease

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Acute graft-versus-host disease (aGVHD) is a life-threatening complication of allogeneic hematopoietic cell transplantation (allo-HCT), for which therapeutic options are limited. Strategies to promote intestinal tissue tolerance during aGVHD may improve patient outcomes. Using single-cell RNA sequencing, we identified a lipocalin-2 (LCN2)-expressing neutrophil population in mice with intestinal aGVHD. Transfer of LCN2-overexpressing neutrophils or treatment with recombinant LCN2 reduced aGVHD severity, whereas the lack of epithelial or hematopoietic LCN2 enhanced aGVHD severity and caused microbiome alterations. Mechanistically, LCN2 induced insulin-like growth factor 1 receptor (IGF-1R) signaling in macrophages through the LCN2 receptor SLC22A17, which increased interleukin-10 (IL-10) production and reduced major histocompatibility complex class II (MHCII) expression. Transfer of LCN2-pretreated macrophages reduced aGVHD severity but did not reduce graft-versus-leukemia effects. Furthermore, LCN2 expression correlated with IL-10 expression in intestinal biopsies in multiple cohorts of patients with aGVHD, and LCN2 induced IGF-1R signaling in human macrophages. Collectively, we identified a LCN2-expressing intestinal neutrophil population that reduced aGVHD severity by decreasing MHCII expression and increasing IL-10 production in macrophages. This work provides the foundation for administration of LCN2 as a therapeutic approach for aGVHD.

INTRODUCTION

Allogeneic hematopoietic cell transplantation (allo-HCT) is the only curative therapy for many hematological malignancies. Still, its success is limited by acute graft-versus-host disease (aGVHD), infections, and relapse. In particular, patients with corticosteroid-refractory (SR) aGVHD have a dismal prognosis, with a reported 1-year survival rate that ranges from 10 to 38% (1, 2). Now, available therapies for

SR-aGVHD include extracorporeal photopheresis and ruxolitinib (3), but further treatment options are necessary.

The intestinal tract plays an important role during the early stages of aGVHD pathogenesis by activating the immune system through damage-associated molecular patterns (4, 5) and pathogen-associated molecular patterns (PAMPs) (6), which are triggered by conditioning-induced tissue damage. Several endogenous factors induce intestinal

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repair mechanisms to cope with tissue damage, including interleukin-22 (IL-22) (7, 8), R-spondin (9, 10), bile acids (11), and glucagon-like peptide-2 (GLP-2) (12). These factors promote the growth of enterocytes, intestinal stem cells, and Paneth cells, thereby restoring gut integrity.

Furthermore, some intestinal immune cells suppress local immune responses to prevent excessive inflammation as a conserved mechanism to cope with an increased translocation of microbes. Innate lymphoid cells (ILCs) are abundant in mucosal tissues such as the gastrointestinal tract. Group 3 ILCs (ILC3s) have been described to contribute to the immune response to bacteria by secreting antimicrobial peptides and cytokines, such as IL-22, promoting barrier integrity (13, 14). In human studies, infiltration of ILC3 was shown to inversely correlate with aGVHD severity, suggesting a protective role in aGVHD (15, 16). Likewise, infusion of ILC2s reduced aGVHD lethality in mouse models of aGVHD by modulating infiltration of T helper cells and myeloid-derived suppressor cells (MDSCs), thereby improving barrier integrity (17).

Tolerance-inducing effects have also been ascribed to intestinal myeloid cell types, such as macrophages, regulatory dendritic cells, and MDSCs (17, 18). Mechanistically, these cells cause local immunosuppression by surface expression of inhibitory ligands, such as programmed cell death ligand 1 (19), or the production of immunosuppressive enzymes, such as indoleamine 2,3-dioxygenase 1 (20). Other intestinal myeloid cells were shown to be pro-inflammatory and enhance aGVHD. In previous studies, we demonstrated that after allo-HCT, monocytes, and granulocytes (neutrophils) enhanced early aGVHD by releasing tissue-damaging factors such as reactive oxygen species (21, 22). In addition, neutrophils were found to migrate from the inflamed gut mucosa to the mesenteric lymph nodes, presenting antigen to donor T cells (23). Because the neutrophil response depended on PAMPs, depletion of tissue-infiltrating bacteria by immunization against poly-*N*-acetylglucosamine reduced intestinal neutrophil accumulation and aGVHD in mice (24).

To elucidate the role of neutrophils during aGVHD succeeding the conditioning-induced acute tissue damage, we performed single-cell RNA sequencing (scRNA-seq) of neutrophils localized in the intestine on day 21 after allo-HCT. This approach identified multiple neutrophil clusters, with two of these clusters expressing lipocalin-2 (LCN2), a secreted glycoprotein produced by epithelial cells (25) and neutrophils (26). At mucosal barriers, LCN2 acts as an antimicrobial peptide impeding the growth of susceptible microbes by sequestering iron-laden microbial siderophores (27). In this context, LCN2 was reported as an indicator of disease activity in inflammatory bowel diseases (IBDs) (28, 29), and LCN2 deficiency exacerbated colitis in *IL-10*^{-/-} mice (30, 31). Moreover, LCN2's pleiotropic roles have been linked to apoptosis, immune modulation, T cell priming, and tumorigenesis (32). By binding to the surface receptor 24p3R/SLC22A17 mainly expressed by macrophages, LCN2 triggers various cellular responses (33). A better understanding of the anti-inflammatory activity of LCN2 and its side effects would help to translate LCN2 treatment into a clinical trial for patients with aGVHD.

RESULTS

scRNA-seq identifies a subset of LCN2⁺ neutrophils in the ileum of mice during aGVHD

Our previous work demonstrated a pro-inflammatory role of neutrophils during early aGVHD (21, 23, 24). To characterize neutrophils

residing in the intestinal tract during later stages of aGVHD at high resolution, we performed a scRNA-seq-based analysis of lamina propria neutrophils from the terminal ileum on day 21 after allo-HCT. We analyzed sorted donor-derived neutrophils as defined by the major histocompatibility complex class I marker, H2-K^b. Cluster analysis with RaceID3 (34) revealed that neutrophils in the ileum did not constitute a homogeneous population, but segregated into multiple distinct states represented by individual clusters. We identified two neutrophil clusters (clusters 9 and 12) with a high expression of major histocompatibility complex class II (MHCII)-encoding genes (Fig. 1A and fig. S1, A and B) that may contribute to antigen presentation to donor T cells, which we have previously described (23). A neutrophil population with low MHCII transcription was found in clusters 6 and 13. This neutrophil subtype was mainly present in mice that had developed aGVHD compared with bone marrow (BM)-only controls (Fig. 1, A and B). Neutrophils in clusters 6 and 13 were characterized by the expression of *Lcn2*. This population was distinct from the MHCII⁺ population for transcription of multiple genes, namely, *S100a8*, *S100a9*, *Retnlg*, *Mmp8*, and *Lrg1* (Fig. 1C). In contrast, the MHCII⁺ population was characterized by increased transcription of multiple MHCII genes and cathepsins (e.g., *Ctss*) (Fig. 1C).

Clustering according to the transcriptional profile allowed us to distinguish the *Lcn2*⁺ population from all other populations (Fig. 1D). Cluster 6 comprised more cells and differentially expressed *Il-1b* and *Cxcl2* in comparison with cluster 13, which displayed higher expression of *Lcn2* and up-regulated genes related to secondary granules (*Camp*, *Ltf*, and *Ngp*) (fig. S1, C and D). Both clusters expressed *Cebpb* and had low expression of *Spi1*, indicating that *Lcn2*⁺ neutrophils represent a mature neutrophil population (fig. S1E). LCN2⁺ neutrophils infiltrating the gastrointestinal tract exhibited an aged phenotype (fig. S1F) and were mainly derived from donor hematopoietic cells as indicated by the neutrophil chimerism of 82 to 94% (fig. S1G). Previous reports on the inhibitory effect of LCN2 on bacterial iron acquisition (27) and its role in IBD (28, 30) motivated us to focus on the functional characterization of the *Lcn2*⁺ neutrophil population in the context of aGVHD. To elucidate the kinetics of *Lcn2* transcription in the small intestine during aGVHD in mice, we analyzed expression by quantitative polymerase chain reaction (qPCR). *Lcn2* mRNA in the ileum was increased on days 7 and 14 after allo-HCT, compared with mice that underwent syngeneic HCT (syn-HCT) (Fig. 1E). LCN2 protein was detected in Ly6G⁺ neutrophils in the ileum of mice at day 21 after allo-HCT (Fig. 1F). LCN2 expression in blood neutrophils was significantly higher in allo-HCT than syn-HCT recipients on day 21 (*P* < 0.0001, Fig. 1G). In addition, LCN2 expression in neutrophils infiltrating the small intestine and colon epithelium on day 14 after transplantation was higher in mice undergoing allo-HCT than mice undergoing syn-HCT (Fig. 1, H and I). LCN2 expression was increased in non-neutrophil myeloid cells only in the small intestine epithelium on day 14 (fig. S1, H and I). Furthermore, we found a high expression of LCN2 in neutrophils infiltrating the colon epithelium of mice in a dextran sodium sulfate (DSS)-induced colitis model (fig. S1J). Because LCN2 can also be secreted by intestinal epithelial cells, we next analyzed its abundance in the stool. The concentration of LCN2 was increased in the stool of allo-HCT mice compared with naïve mice and with mice that underwent syn-HCT on days 7 and 21 after allo-HCT in the animal facility in Freiburg (Fig. 1J). LCN2 in the stool was low in germ-free and antibiotic-treated mice but increased in the stool from days 1 to 8

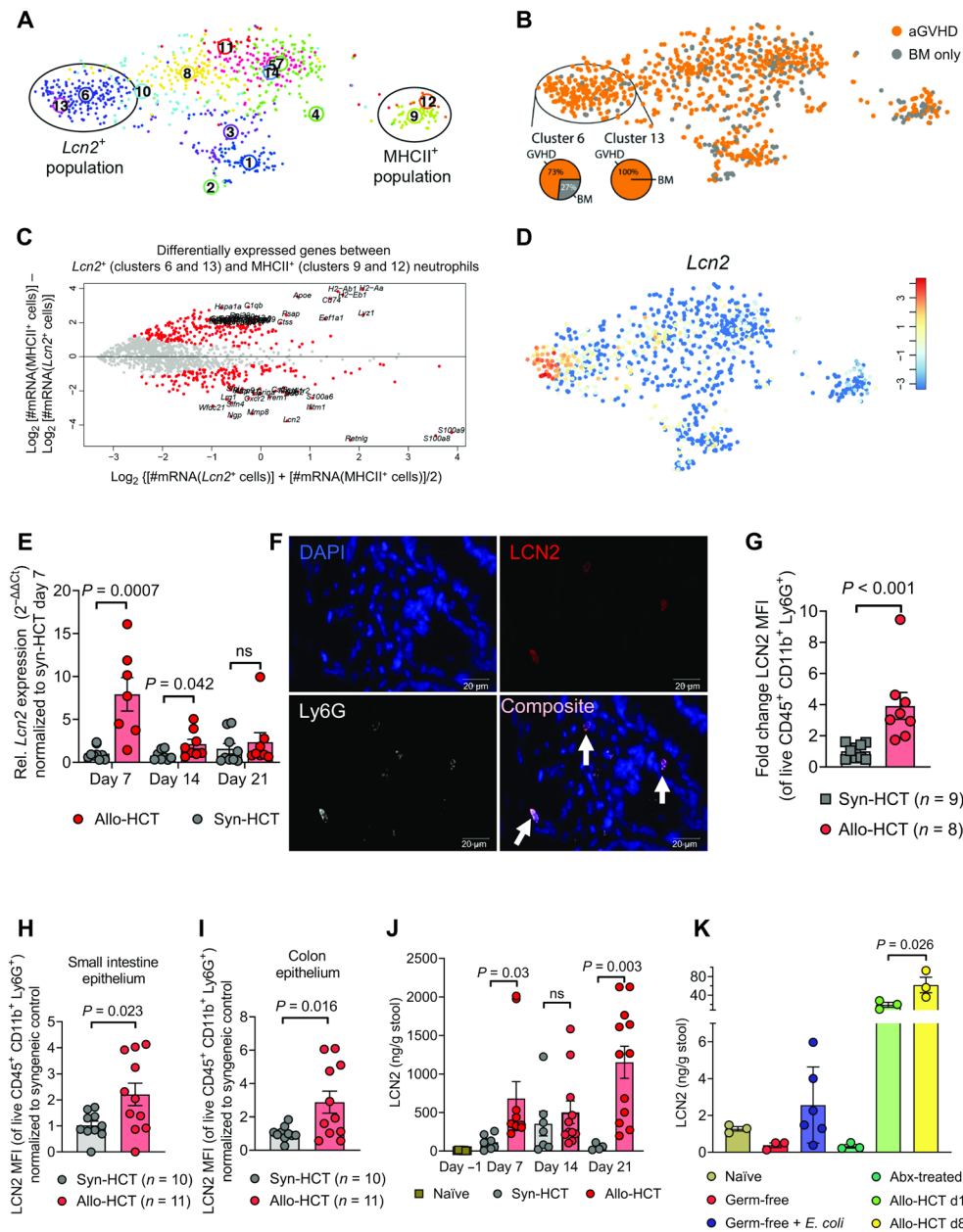


Fig. 1. scRNA-seq identifies a LCN2⁺ neutrophil subtype in the ileum of mice developing aGVHD. (A) CD45⁺H2-K^b+CD11b⁺Ly6G⁺ neutrophils from mice with aGVHD (BM + T cells) or control mice (BM only) 21 days after allo-HCT were analyzed by scRNA-seq. Shown is a t-distributed stochastic neighbor embedding (t-SNE) map highlighting RaceID3 clusters. (B) Shown is a t-SNE map of sorted cells (aGVHD, orange; BM only, gray). Cluster 6 contains 73%, and cluster 13 contains 100% neutrophils from aGVHD mice, after correcting for unequal cell numbers between conditions. A total of 697 cells from aGVHD mice and 226 cells from BM-only mice were analyzed. (C) Shown is an MA plot of differentially expressed genes between the MHCII⁺ (top) and *Lcn2*⁺ (bottom) populations. Genes with significant fold changes between the conditions (Benjamini-Hochberg adjusted $P < 0.05$ and log₂ fold change > 1.9) are shown. (D) Shown is a t-SNE map depicting expression of *Lcn2*. The color scale indicates log₂-normalized expression. (E) Shown is qPCR-based quantification of *Lcn2* mRNA in ileum of allo-HCT recipients (C57BL/6 into BALB/c) and syn-HCT recipients (BALB/c into BALB/c) at the indicated time points, $n = 7$ to 9 mice per group. Statistical analysis by Mann-Whitney U test. ns, not significant; Rel., relative. (F) Representative immunofluorescence staining for LCN2 and Ly6G is shown for frozen tissue sections isolated from ileal lamina propria on day 21 after allo-HCT (C57BL/6 into BALB/c). Arrows indicate double-positive cells. (G) Shown is relative LCN2 protein expression of CD45⁺CD11b⁺Ly6G⁺ neutrophils in the blood of recipients on day 21 after allo-HCT (C57BL/6 into BALB/c) or syn-HCT (BALB/c into BALB/c), $n = 8$ or 9 mice per group. Statistical analysis with Mann-Whitney U test. MFI, mean fluorescence intensity. (H and I) Shown is LCN2 expression of CD45⁺CD11b⁺Ly6G⁺ neutrophils isolated from the epithelium of small intestine (H) or colon (I) of recipients on day 14 after allo-HCT (C57BL/6 into BALB/c) or syn-HCT (BALB/c into BALB/c), $n = 10$ or 11 mice per group. Statistical analysis with unpaired t test. (J) LCN2 (nanograms per gram of stool) was measured by enzyme-linked immunosorbent assay in stool of recipients after allo-HCT (C57BL/6 into BALB/c) and syn-HCT (BALB/c into BALB/c) at indicated time points, $n = 4$ to 12 per group, or untreated BALB/c, $n = 8$ per group. Statistical analysis by Kruskal-Wallis test with Dunn's multiple comparisons test. (K) LCN2 (nanograms per gram of stool) was measured in mice with different microbial colonization. Statistical analysis by Kruskal-Wallis test with Dunn's multiple comparison correction. GF, germ-free; naïve, conventional (specific pathogen-free); Abx, antibiotic-treated; allo-HCT, BALB/c mice that received C57BL/6 BM and Tc, sampled at the indicated time points after transplantation.

after allo-HCT in mice kept in a different animal facility (San Diego), supporting the data acquired in Freiburg (Fig. 1K).

LCN2^{-/-} donor BM transfer exacerbates aGVHD

To clarify the contribution of donor neutrophil-derived LCN2 on the bacterial diversity in the intestinal lumen, we performed 16S ribosomal RNA (rRNA) gene sequencing in mice that had received wild-type BM and wild-type T cells compared with LCN2^{-/-} BM and wild-type T cells. First, we analyzed the effect of allo-HCT on microbial composition by performing nonmetric multidimensional scaling (NMDS) analysis, where we compared samples taken before

transplant (untreated) with samples from animals that received wild-type or LCN2^{-/-} BM during allo-HCT. As expected, untreated versus allo-HCT clustered differently; however, we did not observe differences between the aGVHD groups (Fig. 2A). We then determined the abundance of bacteria on the phylum level in the wild-type donor group versus the LCN2^{-/-} donor group (Fig. 2B) and found that, within the phylum *Firmicutes*, the class *Clostridia* ($P = 0.027$, Fig. 2C) and the family *Lachnospiraceae* ($P = 0.022$, Fig. 2D) were significantly reduced in the LCN2^{-/-} donor group. Loss of *Firmicutes*, specifically several *Clostridia* species, has previously been linked to more severe aGVHD (35). In particular, the family *Lachnospiraceae* has been

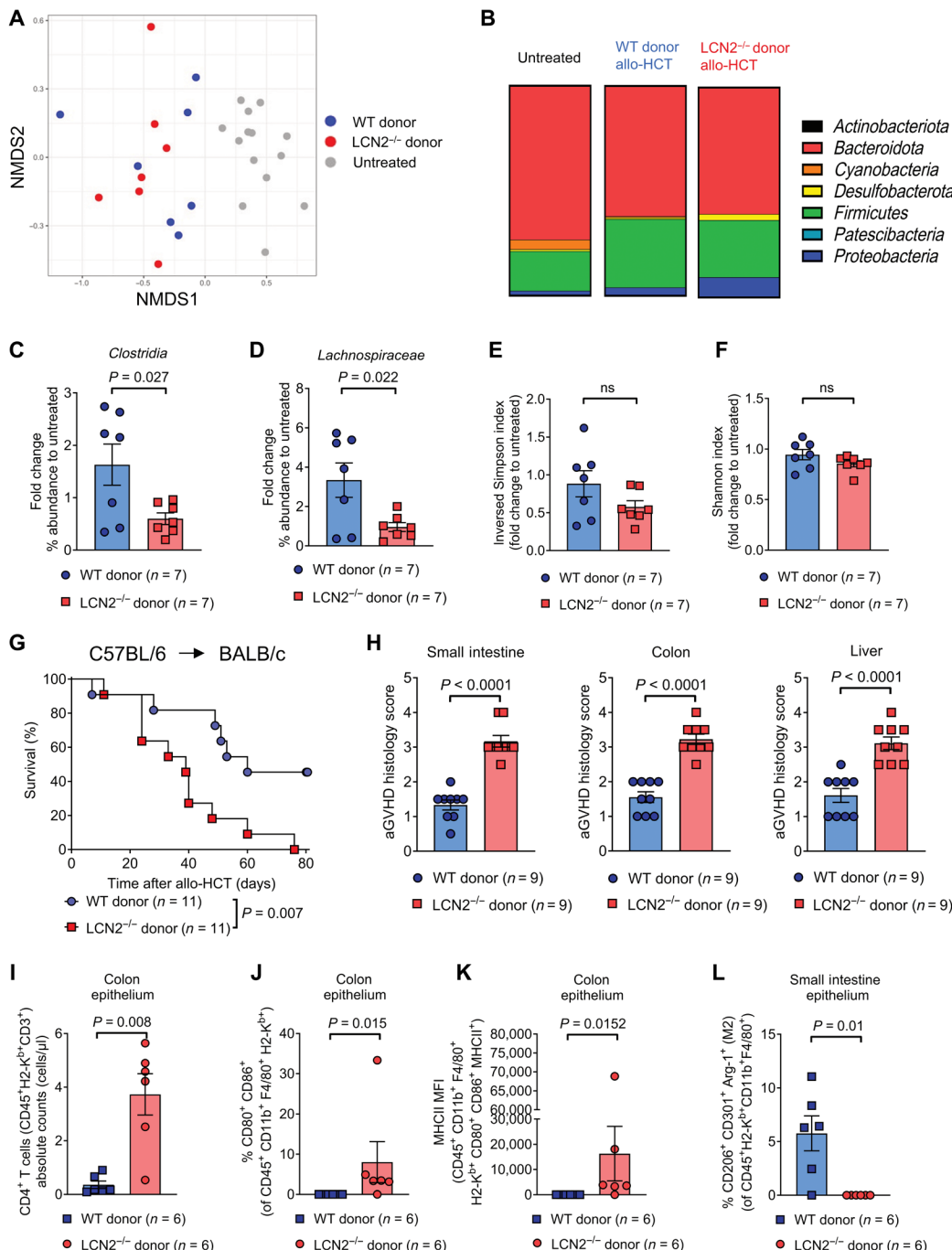


Fig. 2. Loss of hematopoietic LCN2 affects the intestinal microbial composition and exacerbates aGVHD. (A to F) Microbiome analysis was performed by 16S rRNA gene sequencing. Stool samples were collected from BALB/c on day -1 before allo-HCT (untreated) or from recipients on day 14 after allo-HCT. All received C57BL/6 wild-type T cells with wild-type BM (WT donor) or wild-type T cells with LCN2^{-/-} BM (LCN2^{-/-} donor).

(A) NMDS analysis is shown. (B) Shown is a simplified overview of bacteria taxonomy and respective relative abundance at the phylum level. (C and D) Shown is the change of relative abundance after allo-HCT of *Clostridia* (class) (C) and *Lachnospiraceae* (family) (D). Microbial abundance was normalized to pre-transplant abundance, paired analysis, $n = 7$ per group. (E and F) Microbial diversity was represented by inversed Simpson index (E) and Shannon index (F), $n = 7$ animals per group. P values in (C) to (E) were calculated using unpaired t test. (G to L) Shown is the analysis of BALB/c recipients that received wild-type T cells with wild-type BM (WT donor) or wild-type T cells with LCN2^{-/-} BM (LCN2^{-/-} donor). (G) Percent survival is shown, $n = 11$ animals per group, analyzed by Mantel-Cox test. (H) aGVHD histopathology scores are shown for small intestines, colons, and livers isolated on day 14 after allo-HCT, $n = 9$ per group, statistically analyzed by Mann-Whitney U test. (I) Shown is the absolute number of CD4⁺ donor T cells infiltrating colon epithelium on day 14 after allo-HCT, $n = 6$ mice per group. (J) Shown is the percent CD80⁺ CD86⁺ donor (H2-K^b) macrophages (CD11b⁺F4/80⁺) infiltrating colon epithelium on day 14 after allo-HCT, $n = 6$ mice per group. (K) MHCI MFI was quantified for donor (H2-K^b) macrophages (CD11b⁺F4/80⁺) isolated from colon epithelium on day 14 after allo-HCT, $n = 6$ mice per group. (L) Shown is the percent CD206⁺ CD301⁺ arginase-1⁺ donor (H2-K^b) macrophages (CD11b⁺F4/80⁺) isolated from small intestine epithelium on day 14 after allo-HCT, $n = 6$ mice per group. Statistical analysis in (H) to (L) was performed by Mann-Whitney U test.

associated with protection against aGVHD (36), and *Clostridia*-derived butyrate was described to reduce intestinal epithelial cell damage (37). We observed no significant differences with respect to overall microbial diversity between the wild-type and LCN2^{-/-} donor groups ($P = 0.1243$, Fig. 2E; $P = 0.1567$, Fig. 2F).

To clarify the functional role of LCN2 in the donor compartment, we analyzed the immune cells infiltrating the gastrointestinal tract from mice receiving wild-type versus LCN2^{-/-} BM during allo-HCT. LCN2^{-/-} BM recipients developed more severe aGVHD, with a higher mortality than wild-type BM recipients (Fig. 2G). The more aggressive disease course was concurrent with higher aGVHD histopathology scores (Fig. 2H). Because these results do not exclude a potential role of hematopoietic cells other than neutrophils, we also analyzed infiltration of cells in the gastrointestinal tract. We found that the absolute numbers of donor CD4⁺ T cells were increased in the colon epithelium of LCN2^{-/-} BM recipients (Fig. 2I). Furthermore, the percentage of CD80⁺/CD86⁺ donor macrophages present in the colon epithelium was increased in mice that received LCN2^{-/-} BM (Fig. 2J). Macrophages in the LCN2^{-/-} BM group also expressed more MHCII, consistent with macrophage activation, compared with mice receiving wild-type BM (Fig. 2K). Conversely, the frequency of anti-inflammatory CD206⁺CD301⁺arginase-1⁺ (M2) macrophages in the small intestine was reduced in the LCN2^{-/-} BM group compared with mice receiving wild-type BM (Fig. 2L). These findings indicate that lack of LCN2 in the donor hematopoietic system leads to a reduction of *Lachnospiraceae* and *Clostridia*, which belong to *Firmicutes*, and a more aggressive aGVHD course in the recipient mice.

Transfer of LCN2-overexpressing neutrophils or LCN2 treatment reduces aGVHD severity

To complement the loss-of-function experiments using LCN2^{-/-} BM, we next adopted a gain-of-function approach. Therefore, we used the previously reported neutrophil progenitor cell line Hoxb8, which differentiates into neutrophils upon estrogen withdrawal (21). Transduction of Hoxb8 cells with a LCN2 overexpressing vector led to increased LCN2 protein in the homogenized cells (fig. S2, A and B), *Lcn2* RNA expression (fig. S2C), and LCN2 protein secretion (fig. S2D). In agreement with our data, the secretion of LCN2 by neutrophils had been shown before (26, 38). To evaluate the functional role of LCN2⁺ neutrophils in aGVHD, BALB/c recipients underwent allo-HCT, followed by transfer of LCN2-overexpressing Hoxb8 progenitor cells or empty vector control cells. Because our scRNA-seq dataset had identified *Lcn2*⁺ cells on day 21 after allo-HCT (Fig. 1A), we chose to perform the transfer at this time point to increase the number of LCN2⁺ neutrophils in late-stage aGVHD. Mice receiving LCN2-overexpressing Hoxb8 neutrophils exhibited improved survival after allo-HCT compared with empty vector Hoxb8 neutrophil transfer (Fig. 3A). In addition, histopathology scoring of the liver, small intestines, and colon revealed reduced aGVHD severity in recipients of LCN2-overexpressing Hoxb8 neutrophils, compared with mice receiving control Hoxb8 neutrophils (Fig. 3B). Hoxb8 neutrophils infiltrated the gastrointestinal tract starting at 2 hours after transplantation (fig. S2E) and were still abundant in colon and small intestine at 20 hours after transplantation (fig. S2, F to H).

To test whether administration of recombinant LCN2 was sufficient to affect aGVHD severity, we treated mice with LCN2 at a dosage of 3 μ g intraperitoneally on day 7 after transplantation to counterbalance the decline of endogenous LCN2 that we observed after this time point (Fig. 1E). LCN2 treatment improved the survival of mice

undergoing allo-HCT compared with mice receiving vehicle treatment (Fig. 3C). In addition, LCN2 treatment resulted in lower aGVHD histopathology scores of the liver, small intestine, colon, and skin compared with mice receiving vehicle treatment (Fig. 3D). To analyze the immunological microenvironment of the gastrointestinal tract, we performed flow cytometry-based analysis of infiltrating immune cells. We found that, on day 14 after allo-HCT, the percentage of T helper 1 (T_H1)-polarized T cells declined (Fig. 3E), whereas T_H2-polarized T cells in the small intestine lamina propria increased in mice treated with LCN2 as compared with vehicle-treated mice (Fig. 3F). The frequency of M2 macrophages was also increased in the epithelium and lamina propria of the colon (Fig. 3, G and H), indicating a less inflammatory environment in the gastrointestinal tract of LCN2-treated recipients. Treatment with LCN2 had no effect on the neutrophil population infiltrating the small intestine and colon on day 14 after allo-HCT (fig. S3, A to E).

16S rRNA sequencing revealed differential clustering before transplant compared to after transplant (fig. S4A) without changes of major phyla (fig. S4B). Alpha diversity as measured by inverted Simpson, Shannon, and Simpson Index decreased after allo-HCT; however, alpha diversity did not change upon treatment with LCN2 (fig. S4, C to E). Consistent with our previous data, we found that, within the phylum *Firmicutes*, the class *Clostridia* (fig. S4F) was increased in LCN2-treated allo-HCT recipients when comparing relative abundance before transplant with relative abundance on day 14, whereas the order *Lachnospirales* (fig. S4G) and the genus *Ruminococcus* (fig. S4H) of the same class remained unchanged. Decontamination of allo-HCT recipients with an antibiotic cocktail did not affect the protective effects of LCN2 administration (fig. S5). To test the stability of our findings, we probed the effect of LCN2 in a second major mismatch aGVHD model (BALB/c into C57BL/6). Likewise, mice receiving LCN2 after allo-HCT were also protected with respect to survival (Fig. 3I) and aGVHD histopathology scores of the liver, small intestine, and colon, compared with mice receiving vehicle treatment (Fig. 3J). The protective effect of LCN2 treatment in the two independent aGVHD models supports the concept that LCN2-mediated immunomodulatory activity is model independent.

LCN2 decreases the expression of genes involved in antigen processing and presentation

To analyze the impact of LCN2 on immune cells in a well-controlled system, we exposed different cell types to LCN2 in vitro. We found no impact of LCN2 on T cell proliferation (fig. S6, A and B) or on neutrophil activation in vitro (fig. S6, C to F). Lipopolysaccharide (LPS) treatment induced *Lcn2* expression in BM neutrophils in vitro (fig. S6G). In addition, we found that LCN2 induced *Arg1* expression in naïve macrophages and anti-inflammatory macrophages but not in pro-inflammatory macrophages (fig. S6, H to J).

Because of the profound impact of LCN2 on macrophage populations in vivo and in vitro, we next measured the expression of the LCN2 receptor SLC22A17 on various antigen-presenting cells. In line with gene signatures described in the expression library of the ImmGen Consortium (39), *Slc22a17* mRNA was strongly expressed on naïve and IL-4-polarized macrophages (Fig. 4A), which could be confirmed at the protein level by Western blot analysis on naïve macrophages (Fig. 4B). Next, we analyzed the impact of LCN2 on transcriptional changes in macrophages in vitro. We generated BM-derived macrophages (BMDMs) that were differentiated in the presence or absence of LCN2. IL-4 treatment of LCN2-primed BMDMs

resulted in reduced expression of multiple genes involved in antigen processing and presentation, including MHCII (Fig. 4C). Consistently, we found that expression of *Ciita*, a transcriptional activator of MHCII, decreased in macrophages upon LCN2 exposure in vitro (Fig. 4D). To validate whether these results were relevant in vivo, we isolated intestinal immune cells from LCN2-treated allo-HCT recipients. MHCII expression was reduced on the recipient (H2-K^{d+}, BALB/c) and donor (H2-K^{b+}, C57BL/6) macrophages located in the colon on day 14 after allo-HCT upon LCN2 treatment (Fig. 4, E and F). Similarly, MHCII expression was reduced on donor (H2-K^{b+}) CD11c⁺CD11b⁺ dendritic cells located in the small intestine on day 14 after allo-HCT upon LCN2 treatment (Fig. 4G). To test the functional impact of LCN2 on anti-inflammatory macrophages, we transferred in vitro

LCN2-exposed or control macrophages into allo-HCT recipients on day 1 after transplantation. Mice receiving LCN2-primed macrophages exhibited prolonged survival compared with mice receiving control macrophages (Fig. 4H). Improved survival was accompanied by lower clinical aGVHD severity scores (fig. S7) and reduced aGVHD histopathology scores (Fig. 4I).

Non-hematopoietic cell-derived LCN2 protects from aGVHD and microbiome disruptions

We next investigated the role of recipient-derived LCN2 in aGVHD using LCN2^{-/-} mice as recipients. To clarify whether any intrinsic differences between wild-type and LCN2^{-/-} recipients were detectable, we measured caecum length, colon length, and other parameters.

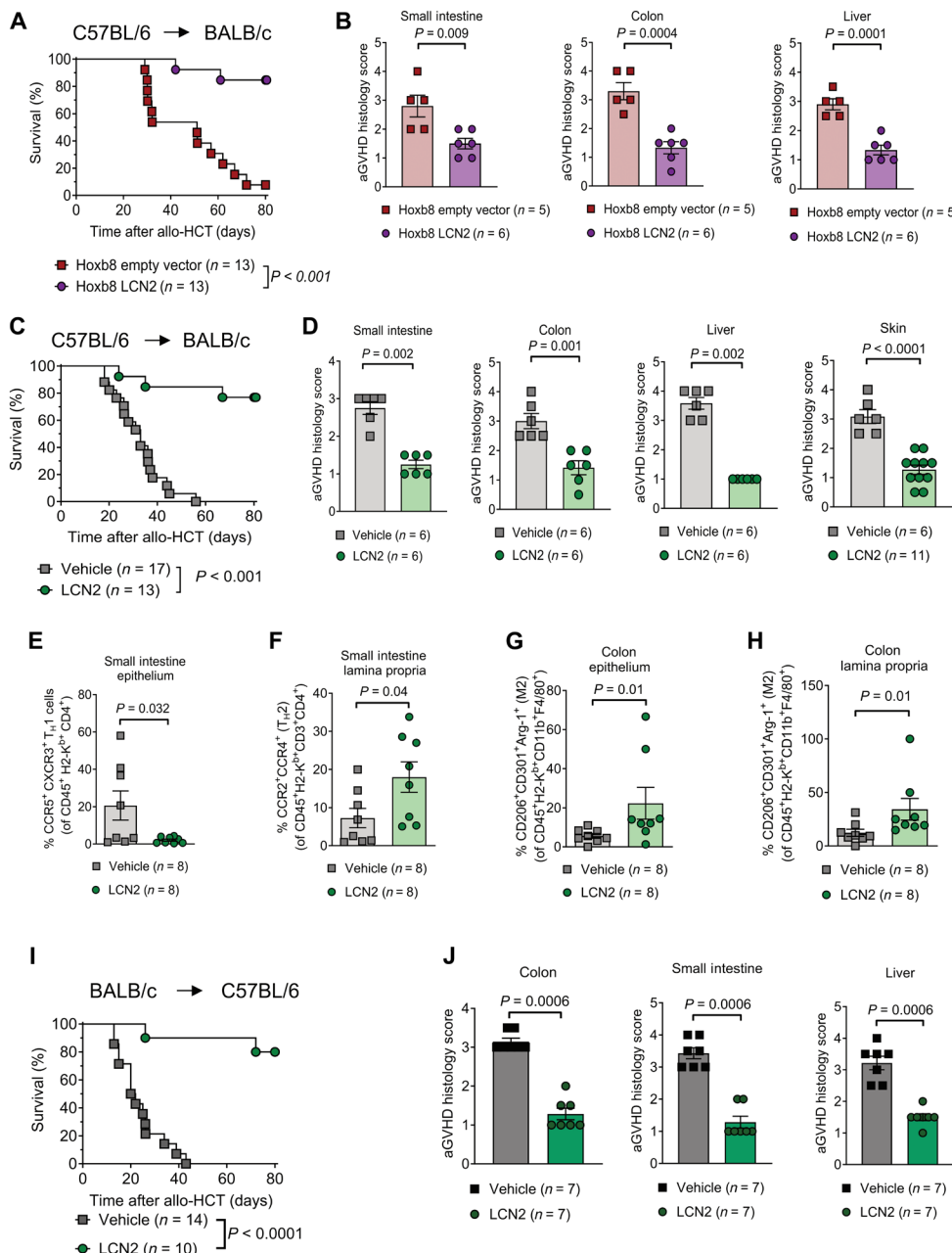


Fig. 3. LCN2 overexpression in neutrophils or LCN2 treatment reduce aGVHD severity. (A and B) Shown are data from an allo-HCT aGVHD model (C57BL/6 into BALB/c) with adoptive transfer of donor-type Hoxb8 cells on day 21. Hoxb8 neutrophils were transduced with control plasmid (Hoxb8 empty vector) or LCN2-overexpression vector (Hoxb8 LCN2). (A) Shown is percent survival $n = 13$ per group; statistical analysis was performed using a Mantel-Cox test. (B) aGVHD histopathology scores are shown for the small intestine, colon, and liver on day 33 after allo-HCT, $n = 5$ or 6 per group. Statistical analysis by unpaired t test. (C to H) Allo-HCT (C57BL/6 into BALB/c) recipients were treated with 3 μ g of recombinant murine LCN2 (rmLCN2) or vehicle on day 7. (C) Percent survival of recipients is shown. Pooled data, $n = 13$ to 17 per group. Statistical analysis was performed using a Mantel-Cox test. (D) aGVHD histopathology scores are shown for small intestine, colon, liver, and skin isolated on day 14 after allo-HCT, $n = 6$ to 11 mice per group. Statistical analysis by Mann-Whitney U test or unpaired t test. (E) Shown is the percent CCR5⁺CXCR3⁺T_H1 cells of all donor (H2-K^{b+}) CD45⁺CD3⁺CD4⁺ T cells in ileal epithelium on day 14 after allo-HCT, $n = 8$ mice per group. (F) Percent CCR2⁺CCR4⁺T_H2 T cells of all donor (H2-K^{b+}) CD45⁺CD3⁺CD4⁺ T cells in ileal lamina propria on day 14 after allo-HCT, $n = 8$ mice per group. Statistical analysis in (E) and (F) by unpaired t test. (G and H) Percent CD206⁺CD301⁺arginase-1⁺ macrophages of all donor (H2-K^{b+}) macrophages (CD11b⁺F4/80⁺) in colon epithelium (G) or lamina propria (H) on day 14 after allo-HCT, $n = 8$ mice per group. Statistical analysis in (G) and (H) by Mann-Whitney U test. (I and J) allo-HCT major mismatch model (BALB/c into C57BL/6) recipients were treated with 3 μ g of rmLCN2 intraperitoneally or vehicle on day 7. (I) Percent survival of C57BL/6 recipients is shown. Pooled data, $n = 10$ to 14 per group. Statistical analysis was performed using a Mantel-Cox test. (J) aGVHD histopathology scores are shown for small intestine, colon, and liver isolated on day 14 after allo-HCT, $n = 7$ mice per group. Statistical analysis by Mann-Whitney U test.

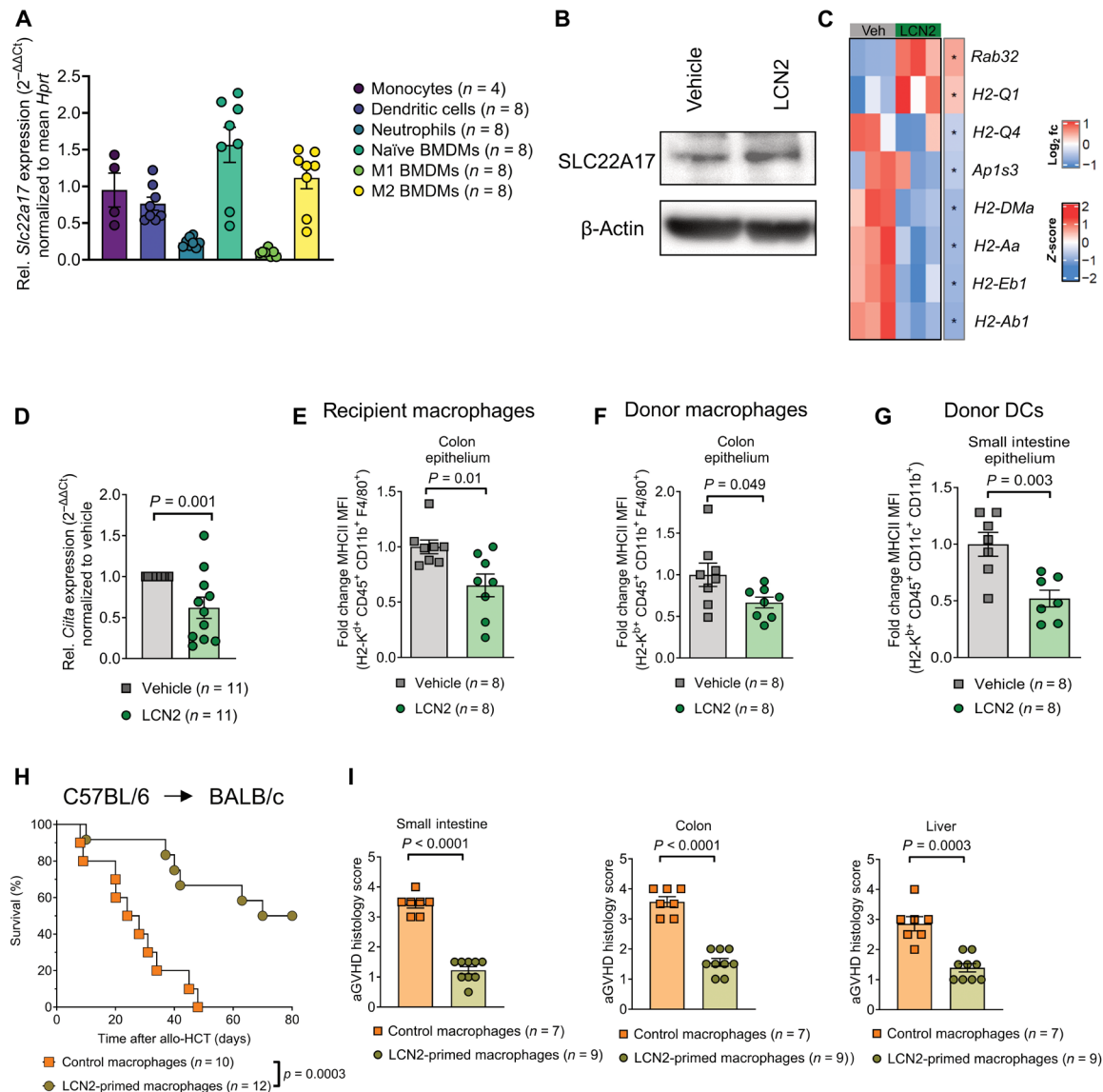


Fig. 4. LCN2 reduces expression of genes involved in antigen processing and presentation. (A) Shown is qPCR analysis of *Slc22a17* expression in myeloid cells isolated (monocytes and neutrophils) or generated (dendritic cells and macrophages) from BM. Delta cycle threshold (ΔC_t) values were normalized to mean *Hprt* ΔC_t . (B) Shown is a Western blot of SLC22A17 in BMDMs treated with vehicle or rmLCN2 (150 ng/ml). A representative blot from four biological replicates is shown. (C) Shown are differentially expressed genes annotated with "Gene Ontology antigen processing and presentation" generated from microarray data. RNA was isolated from IL-4-stimulated BMDMs exposed to vehicle (Veh) or rmLCN2 (150 ng/ml). Color legend indicates row-wise scaling of normalized intensity (Z-score, black box) and log₂ fold change (log₂ fc; gray box). Asterisks indicate significant regulation (adjusted P value < 0.05). Each column represents a biological replicate, $n = 3$ per group. (D) Shown is qPCR analysis of *Ciita* mRNA measured in LCN2- or vehicle-treated IL-4-stimulated BMDMs, $n = 11$ biological replicates. Statistical analysis by one-sample Wilcoxon test. (E to G) Allo-HCT major mismatch model (C57BL/6 into BALB/c) recipients were treated with LCN2 intraperitoneally or vehicle on day 7 and analyzed on day 14. Pooled data from $n = 8$ per group, statistical analysis by Mann-Whitney U. (E and F) Fold change of MHCII expression (MFI) on recipient (H2-K^b+) (E) and donor (H2-K^b+) (F) macrophages (CD11b⁺F4/80⁺) isolated from colon epithelium. (G) Fold change of MHCII MFI on donor (H2-K^b+) dendritic cells (CD11b⁺CD11c⁺) isolated from ileal epithelium. (H and I) Allo-HCT (C57BL/6 into BALB/c) recipients were subjected to adoptive transfer of vehicle- or LCN2-primed, IL-4-stimulated BMDMs on day 1 after transplantation. (H) Shown is percent survival of BALB/c recipients, $n = 10$ to 12 from two independent experiments. Statistical analysis was performed using a Mantel-Cox test. (I) aGVHD histopathology scores are shown for small intestine, colon, and liver isolated day 14 after allo-HCT, $n = 7$ to 9 mice per group. Statistical analysis by Mann-Whitney U test or unpaired t test.

We observed no differences between wild-type and LCN2^{-/-} mice (fig. S8, A to F). Histological analysis of colons, small intestines, livers, and skins of one LCN2^{-/-} mouse cohort also revealed no differences (fig. S8, G and H). This was reproducible in a second cohort of wild-type and LCN2^{-/-} mice (fig. S8I). In addition, LCN2^{-/-} mice did not differ from their wild-type littermates with respect to the relative abundance of phyla and microbial diversity (fig. S8, J to M). After

transplantation with wild-type BM and T cells (BALB/c into C57BL/6), histopathological analysis revealed more severe inflammation and other pathological aGVHD features in the colons and livers of LCN2^{-/-} mice compared with wild-type recipients (Fig. 5, A to D). To assess the impact of recipient LCN2-deficiency on the intestinal microbiome during aGVHD, we performed 16S rRNA sequencing. Analysis of the microbial profiles by NMDS revealed a significant compositional

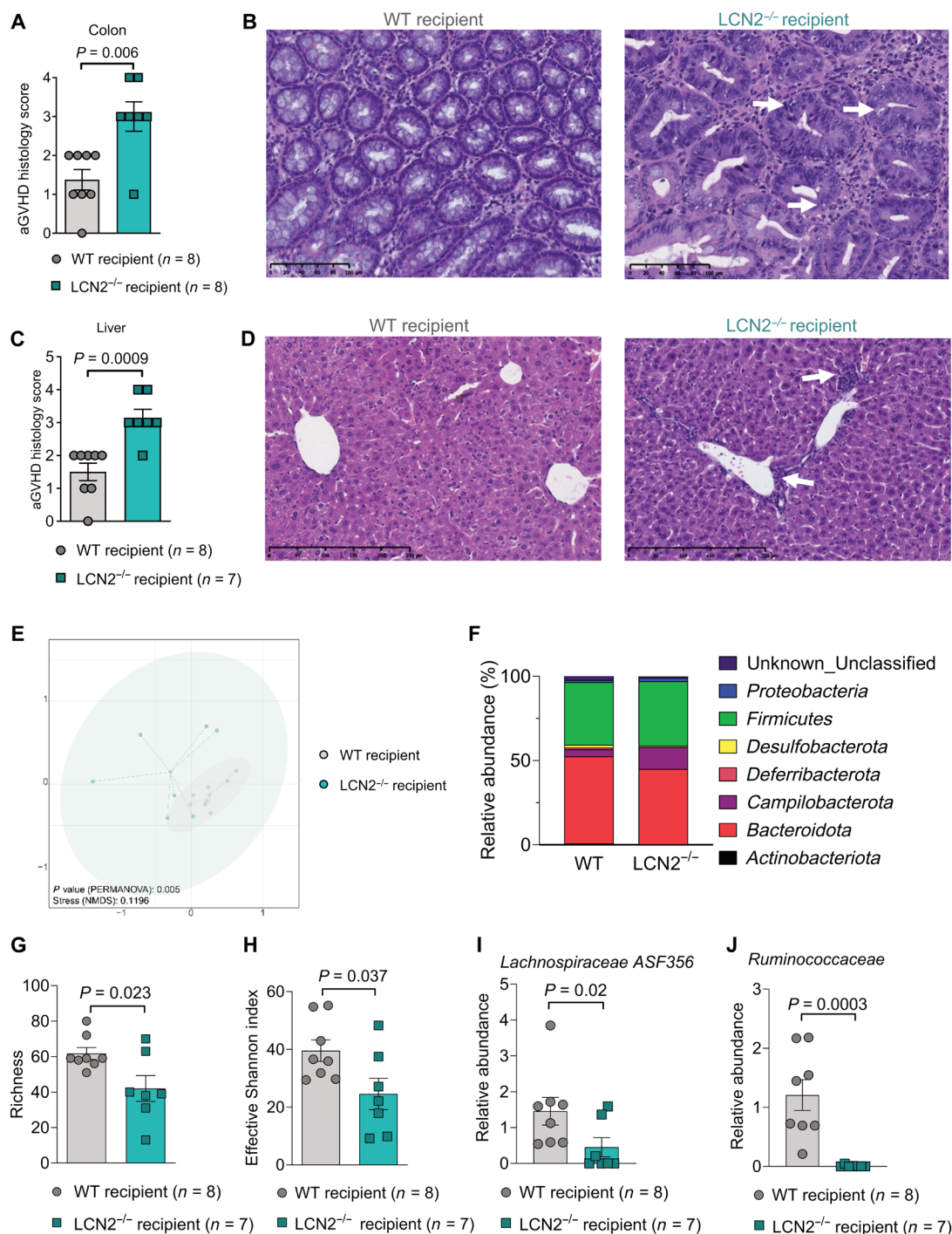


Fig. 5. Non-hematopoietic LCN2 protects against GVHD. Shown is the analysis of major mismatch allo-HCT (BALB/c into C57BL/6) recipients that were wild-type (WT recipient) or LCN2^{-/-} (LCN2^{-/-} recipient), receiving wild-type BM and T cells. (A and B) Shown are histopathological aGVHD scores of the colon (A) and representative hematoxylin and eosin (H&E)-stained sections (B) of colon tissues isolated on day 29. Scale bars, 100 μ m. The white arrow points toward apoptotic cells. (C and D) Shown are histopathological aGVHD severity scores of the liver (C) and representative H&E-stained sections (D) of livers isolated on day 29. Scale bars, 250 μ m. The white arrow points toward infiltrating immune cells surrounding the small bile ducts. Statistical analysis in (A) and (C) was conducted by Mann-Whitney *U* test. (E to J) Shown are results from 16S rRNA gene sequencing. (E) Microbial profiles of wild-type ($n = 8$) and LCN2^{-/-} ($n = 7$) recipient mice are shown on the basis of NMDS analysis. Statistics from permutational dispersion and multivariate analysis of variance (PERMANOVA) are indicated. (F) Shown is the mean relative abundance of phylogenetic taxa at the phylum level in wild-type and LCN2^{-/-} groups. (G and H) Microbial richness (G) and effective Shannon diversity (H) are shown for wild-type and LCN2^{-/-} recipients. *P* values; statistics were calculated using unpaired *t* test. (I and J) Relative abundances of the families *Lachnospiraceae* (I) and *Ruminococcaceae* (J) are shown for wild-type versus LCN2^{-/-} recipients; statistical analysis by Mann-Whitney *U* test.

shift of the fecal microbiota in LCN2 recipients compared with wild-type control littermates ($P = 0.005$, Fig. 5E), which was further confirmed at the phylum level (Fig. 5F). The alpha diversity of groups was assessed at the operational taxonomic unit level by considering effective richness ($P = 0.023$, Fig. 5G) and Shannon effective diversity ($P = 0.037$, Fig. 5H), with both indices being lower in LCN2^{-/-} recipients. LCN2^{-/-} mice exhibited a significant loss of the families *Lachnospiraceae* and *Ruminococcaceae* ($P = 0.02$, Fig. 5I; and $P = 0.0003$, Fig. 5J), both of which have been associated with protection from aGVHD (40, 41). These findings indicate that non-hematopoietic cell-derived LCN2 affects the composition of the gut microbiota in the context of allo-HCT and protects mice from aGVHD.

LCN2 induces IL-10 expression in macrophages

To decipher the mechanism of LCN2-mediated protection from aGVHD, we analyzed expression of anti-inflammatory mediators in LCN2-primed BMDMs. We found that LCN2 treatment induced *IL-10* expression in naïve macrophages in vitro (Fig. 6A). Although wild-type macrophages exhibited reduced MHCII expression upon LCN2 exposure, this effect was absent in *IL-10*^{-/-} macrophages (Fig. 6B), suggesting a connection between IL-10 and MHCII expression. Consistent with a central role of the LCN2–IL-10 axis in protecting from aGVHD, recipients of IL-10-deficient BM were not protected from aGVHD-related mortality by LCN2 treatment (Fig. 6C) and exhibited aGVHD histopathology scores comparable to vehicle-treated mice despite LCN2 treatment (Fig. 6D). These results were validated using an antibody-mediated depletion model (Fig. 6E).

Because IL-10 induction and MHCII down-regulation could influence the graft-versus-leukemia (GVL) effect, we used two previously described mouse leukemia models (12). We observed that the GVL effect was intact in both leukemia models when mice received T cells in terms of survival (Fig. 6, F and G) and reduced leukemic burden (Fig. 6, H and I). The GVL effect was not reduced by treatment with LCN2 in either model (Fig. 6, H and I). Together, these findings indicate that the effect of LCN2 in the context of aGVHD is at least partially mediated by IL-10 (42) and does not interfere with GVL effects.

LCN2 induces IGF-1R signaling in macrophages

To elucidate the downstream signaling pathways affected by LCN2, interferon- γ (IFN- γ) and LPS-activated BMDMs were exposed to LCN2, and a PamChip tyrosine kinase activity array was performed. We observed differential phosphorylation of multiple peptides in macrophages exposed to LCN2 compared with vehicle (Fig. 7A). Analysis of these peptides revealed a list of predicted tyrosine kinases whose activity was modulated by LCN2 treatment. Among these, FGR and insulin-like growth factor 1 receptor (IGF-1R) activation by LCN2 were the top hits (Fig. 7B) and were, therefore, evaluated in further experiments. We validated the increase of IGF-1R activity mediated by autophosphorylation of Tyr^{1161/1165/1166} by Western blot, demonstrating increased IGF-1R phosphorylation in BMDMs exposed to LCN2 compared with vehicle (Fig. 7, C and D). IGF-1R signaling was previously linked to an anti-inflammatory macrophage phenotype (43). Because signal transducer and activator of transcription 6 (STAT6) activation is a central event for the differentiation of naïve macrophages into anti-inflammatory M2 macrophages (44, 45), we thus analyzed STAT6 phosphorylation, finding that LCN2 induced STAT6 phosphorylation in naïve macrophages but not in M1-polarized macrophages (Fig. 7, E and F).

LCN2 is increased in intestinal biopsies of patients with aGVHD

To understand whether our findings in mice could be extrapolated to patients, we next analyzed the expression of LCN2 mRNA in intestinal biopsies from a cohort of patients that had undergone allo-HCT at the University Hospital Regensburg (table S1). LCN2 mRNA was increased in patients with aGVHD grades 2 to 4 compared with patients without aGVHD (Fig. 8A). By implementing a binary logistic regression model, we found that LCN2 expression was dependent on GI-GvHD [Exp(B) = 5.5, 95% confidence interval (CI) = 1.9 to 15.9, $P = 0.001$] but independent of steroid treatment [Exp(B) = 1.2, 95% CI = 0.4 to 3.3, $P = 0.6$] (table S2). Consistently, aGVHD severity correlated with the mRNA expression of LCN2 in patients with aGVHD in this cohort (Fig. 8B). We further validated our findings in a recently reported aGVHD patient cohort from Minnesota (46). RNA-seq data from the Minnesota cohort (46) revealed increased LCN2 expression in the gastrointestinal tracts of patients at aGVHD onset and in patients with SR-aGVHD compared with those in healthy controls (Fig. 8C), which is consistent with increased LCN2 in mice developing aGVHD. LCN2 and *IL-10* expression were analyzed in the Minnesota cohort, allowing for correlative studies of multiple genes. We found that LCN2 expression was correlated with *IL-10* expression in biopsies from patients (Fig. 8D). In samples from a cohort in Freiburg, LCN2 co-localized with CD15⁺ neutrophils in patients with aGVHD (Fig. 8E), and patients with aGVHD grades 2 to 4 (table S3) had higher numbers of LCN2-expressing neutrophils than allo-HCT patients without aGVHD (Fig. 8F). At the time of analysis, all patients had 100% donor chimerism (table S3). In an additional cohort of patients in Innsbruck, human LCN2 in the serum was increased in patients with aGVHD compared with patients undergoing allo-HCT without aGVHD (Fig. 8, G and H, and table S4), with elevated LCN2 expression at clinical onset of aGVHD symptoms (fig. S9A and table S5). In this cohort, LCN2 in the serum significantly decreased in patients that responded to corticosteroids, whereas LCN2 was stable in patients without aGVHD or in patients that were steroid refractory ($P = 0.01$, fig. S9B). To validate the impact of LCN2 on signaling, we next analyzed human monocyte-derived macrophages (hMDMs) exposed to LCN2. We observed that hMDMs expressed the LCN2 receptor SLC22A17 (Fig. 8I). Comparable to mice, IGF-1R phosphorylation increased upon exposure to LCN2 in naïve hMDMs (Fig. 8I). In contrast to naïve hMDMs and pro-inflammatory BMDMs, we observed that LPS- and IFN- γ -exposed hMDMs exhibited no phosphorylation of IGF-1R (Fig. 8I), which may be due to intrinsic differences between mouse and human monocytes. Collectively, our findings demonstrate that LCN2-expressing neutrophils are found in the intestinal tract of mice with aGVHD and that LCN2 is highly up-regulated in the gut of patients with aGVHD. LCN2 induces IGF-1R phosphorylation in murine and human macrophages and leads to reduced MHCII expression, causing a less severe GVHD phenotype (fig. S10).

DISCUSSION

The introduction of scRNA-seq has led to the identification of rare cell types and multiple immune cell subsets previously not detectable (47). Here, scRNA-seq of intestinal neutrophils led to the identification of a LCN2⁺ neutrophil subset that was distinct from other neutrophil clusters on the basis of its mRNA expression profile.

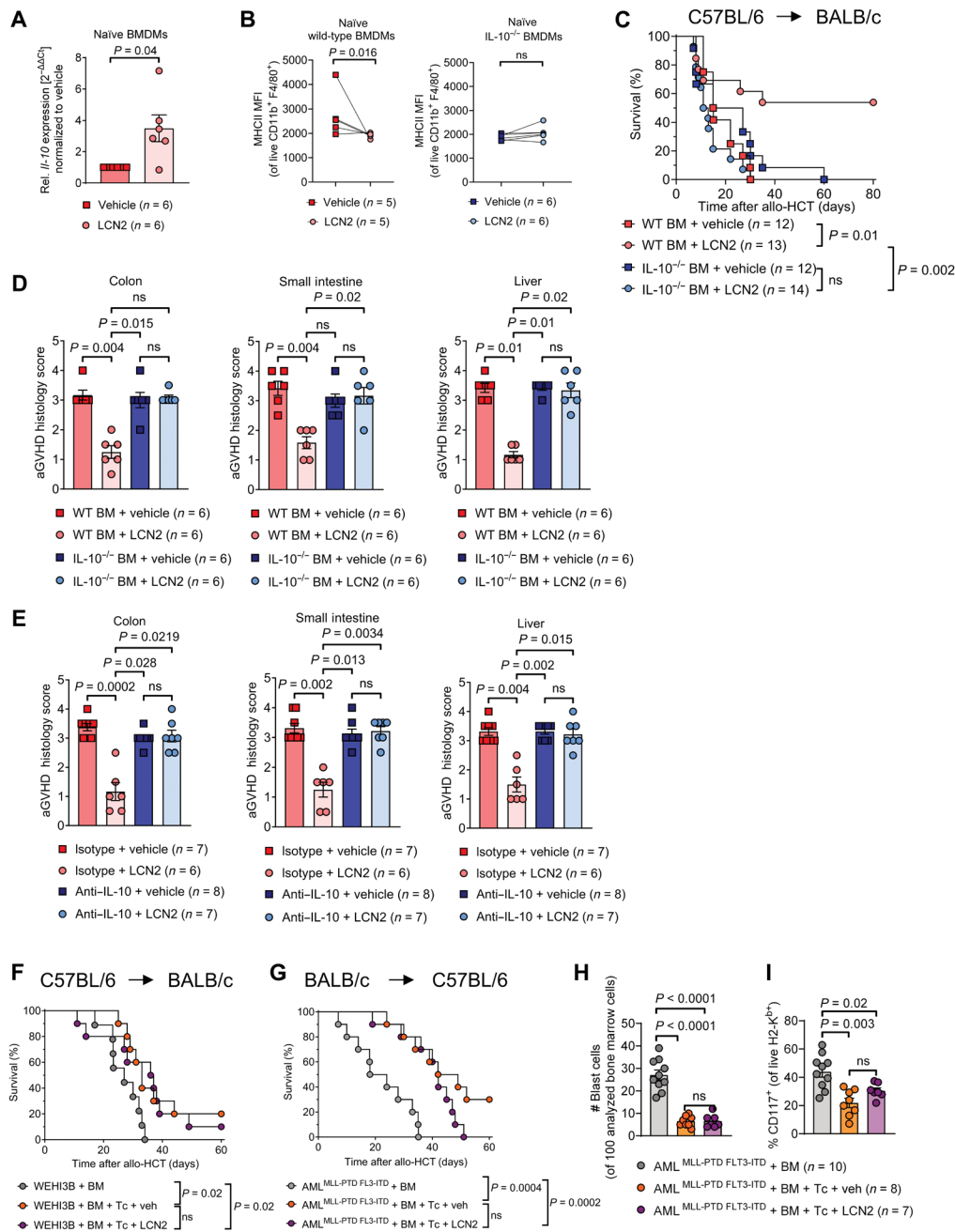


Fig. 6. LCN2-mediated protection requires hematopoietic expression of *IL-10*. (A) Shown is qPCR analysis of *IL-10* expression in LCN2- or vehicle-primed IL-4-treated macrophages. $n = 6$ biological replicates. Statistical analysis by one-sample Wilcoxon test. (B) Shown is the MFI of MHCII in LCN2- or vehicle-primed macrophages from wild-type (left) or IL-10 $^{-/-}$ (right) C57BL/6 mice, relative to vehicle. $n = 5$ or 6 biological replicates. Statistical analysis by Mann-Whitney U test. (C and D) Allo-HCT (C57BL/6 into BALB/c) recipients were transplanted with wild-type BM and wild-type T cells (WT BM) or IL-10 $^{-/-}$ BM and wild-type T cells (IL-10 $^{-/-}$ BM). Recipients were treated with LCN2 or vehicle intraperitoneally on day 7. (C) Percent survival of recipients is shown, $n = 12$ to 14 per group, pooled data. Statistical analysis was performed with a two-sided Mantel-Cox test. (D) aGVHD histopathology scores are shown for small intestines, colons, and livers isolated on day 14. $n = 6$ mice per group. (E) Allo-HCT (C57BL/6 into BALB/c) recipients were injected intraperitoneally with 300 μ g of IL-10 neutralizing antibody (anti-IL-10) or isotype control (isotype) on day 5 after transplantation. Animals treated with LCN2 or vehicle on day 7. aGVHD histopathology scores of small intestines, colons, and livers isolated on day 14. $n = 6$ or 7 mice per group. Statistical analysis in (D) and (E) was performed by Kruskal-Wallis test and Dunn's correction for multiple comparisons. (F) Shown is the percent survival of BALB/c recipients after transfer of syngeneic AML^{WEHI3B} cells and allo-HCT (C57BL/6 into BALB/c). Treatment with LCN2 intraperitoneally occurred on day 7. $n = 10$ animals per group. Statistical analysis was performed with a Mantel-Cox test. (G to I) Recipient mice received transfer of syngeneic AML^{MLL-PTD FLT3-ITD} cells and allo-HCT (BALB/c into C57BL/6). LCN2 was administered intraperitoneally on day 7. $n = 10$ animals per group. (G) Shown is the percent survival of recipients. Statistical analysis was performed with a Mantel-Cox test. (H) Blast cells were counted in May-Grunwald-Giemsa-stained BM smears of recipients on day 19 after transplantation, $n = 7$ to 10 animals. (I) Percent CD117 $^+$ of H2-K b* cells in BM of recipients on day 19 after transplantation, $n = 7$ to 10 animals; statistical analysis in (H) and (I) was conducted with ordinary one-way ANOVA with a Holm-Sidak correction for multiple comparisons.

Although we have previously shown a detrimental role for bulk neutrophils in the context of early aGVHD (21, 23, 48), this LCN2⁺ neutrophil subset had opposing effects and instead ameliorated aGVHD in mice after allo-HCT. Thus, LCN2⁺ neutrophils should be added to the list of immunoregulatory cell types with anti-inflammatory activity in the context of aGVHD, including T regulatory cells (49), type 1 T regulatory cells (50), and MDSCs (18). Because we observed protective effects against aGVHD upon transfer of the LCN2-overexpressing Hoxb8 neutrophil progenitor cells, transfer of third-party LCN2⁺ neutrophils could be a potential therapeutic approach for aGVHD. Furthermore, because the administration of

recombinant LCN2 protected mice from aGVHD, treatment with LCN2 could be an alternative to cell-based therapy. Previously published data showed increased LCN2 secretion from neutrophils isolated from murine blood upon exposure to LPS *in vitro* (38), which we confirmed in BM-derived neutrophils. An additional mechanism of LCN2 induction is the release of IFN- γ and tumor necrosis factor- α (51). These cytokines contribute to aGVHD [reviewed in (2)] and might contribute to an increased LCN2 pool in the gastrointestinal tract. In addition, besides myeloid cells, epithelial and mucosal tissues of the liver and gut are major sources of secreted LCN2. During intestinal infection or inflammation, LCN2 is induced in an

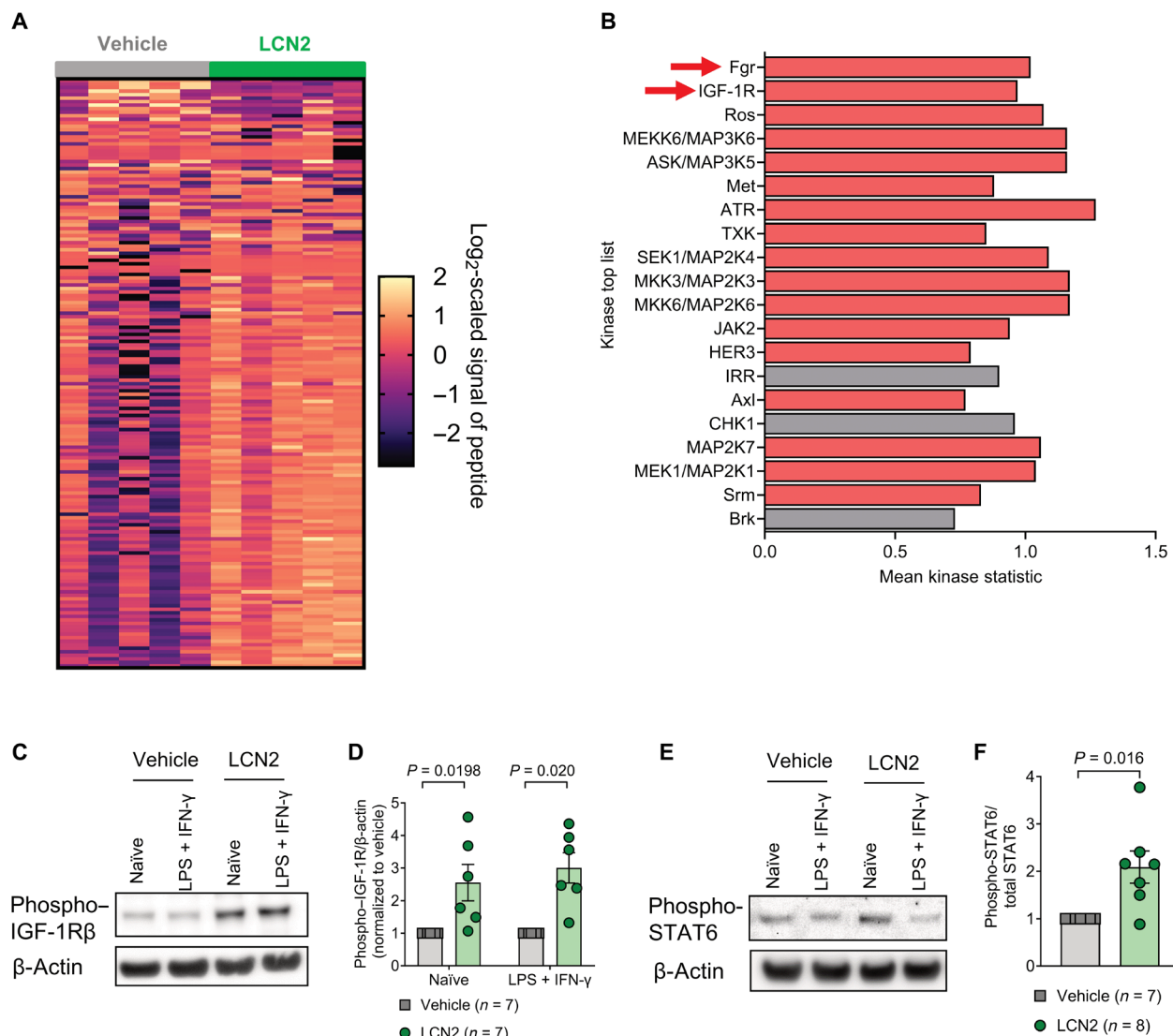


Fig. 7. LCN2 induces IGF-1R signaling in macrophages. (A) Shown is a heatmap from phospho-tyrosine kinase profiling depicting phosphorylation status of 196 peptides in lysates from vehicle- or LCN2-primed, LPS- and IFN- γ -treated BMDMs. Shown is the log₂-scaled signal of peptides (rows), sorted by row mean/clustering or by correlation to factor. (B) Shown is a hit list of kinases potentially differentially active upon LCN2 treatment from lysates described in (A). Red, mean specificity score > 1; gray, mean specificity score < 1 (higher chance of hit by chance). FGR and IGF-1R are highlighted with red arrows. (C to F) Protein from LCN2- or vehicle-primed LPS- and IFN- γ -treated or naive macrophages was subjected to immunoblotting. (C) Representative Western blot for phospho-IGF-1R β . (D) Pooled signal intensity of Western blots for phospho-IGF-1R β . Protein was isolated from LCN2- or vehicle-primed pro-inflammatory or naive macrophages. Data were normalized to the respective vehicle control. Data were pooled from $n = 7$ biological replicates. P values were calculated using two-way ANOVA with Šidák's multiple comparisons correction. (E) Representative Western blot image of phospho-STAT6. (F) Ratio of phospho- to total STAT6 in naive macrophages based on quantification of Western blot images and normalized to β -actin and vehicle control. Statistical analysis was performed by one-sample Wilcoxon test.

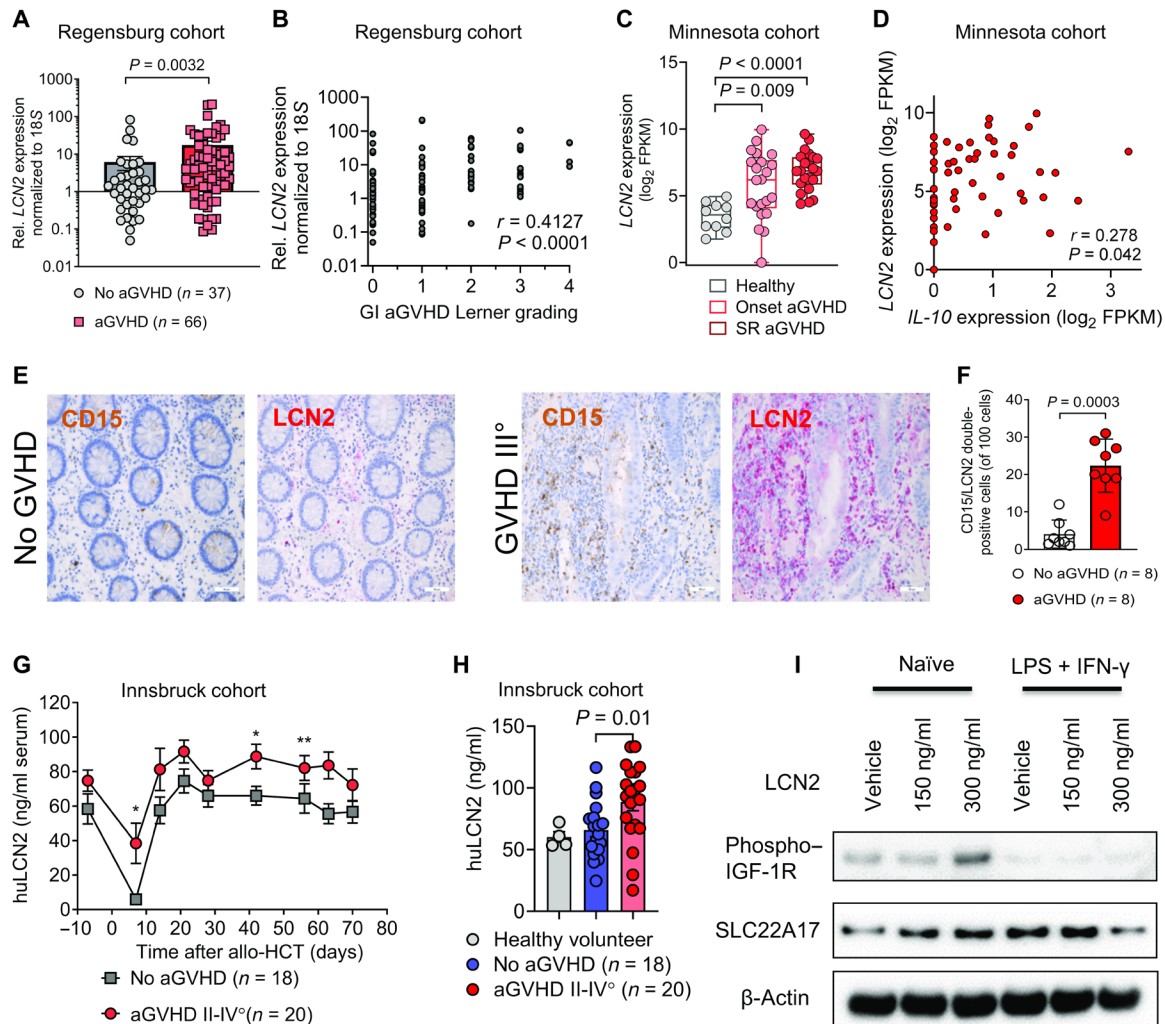


Fig. 8. LCN2 increases in the intestinal tract of patients with aGVHD. (A) Expression of LCN2 RNA was measured in intestinal biopsies on day 30 after allo-HCT. Patients clinically had no aGVHD (no aGVHD, $n = 37$) or grade 2 to 4 aGVHD (aGVHD, $n = 66$). Statistical analysis was performed by Mann-Whitney U test. (B) LCN2 RNA expression values from (A) were plotted against respective GI aGVHD Lerner grade. A Spearman-ranked correlation analysis was performed and tested for statistical significance with t test. (C) Analysis of publicly available RNA-seq data (46) (Minnesota cohort) for expression of LCN2. Healthy, no aGVHD; onset aGVHD; SR-GVHD. Statistical analysis was conducted by unpaired t test. FPKM, fragments per kilobase million. (D) Analysis of Minnesota cohort as in (C) for expression of LCN2 and IL-10. A Spearman-ranked correlation analysis was performed and tested for statistical significance with t test. (E and F) Intestinal biopsies of patients with aGVHD or without aGVHD were stained for CD15 and LCN2 by immunohistochemistry. (E) Representative images for CD15 (neutrophils, brown) and LCN2 (red). Scale bars, 10 μ m. (F) Shown is the quantification of CD15/LCN2 double-positive cells from 100 counted cells per slide. $n = 8$ patients per group. Statistical analysis was performed by Mann-Whitney U test. (G) Human LCN2 (huLCN2) concentrations in serum of patients with ($n = 20$, red circles) and without ($n = 18$, gray squares) aGVHD. Statistical analysis was performed using Holm-Šidák-corrected multiple t tests. (H) Serum LCN2 concentrations were measured on day 42 after allo-HCT in patients with (aGVHD II-IV°, $n = 20$) or without (no aGVHD, $n = 18$) aGVHD and in healthy controls ($n = 4$). Statistical analysis was performed by Mann-Whitney U test. (I) hMDMs from healthy donors were treated with rhLCN2 (150 or 300 ng/ml) and LPS and IFN- γ . Expression of phospho-IGF-1R and SLC22A17 was measured by immunoblot. Shown is a representative blot from one of the six donors.

IL-22-dependent manner and exhibits antimicrobial activity by neutralizing bacterial iron-scavenging siderophores (52–54). In the context of allo-HCT, LCN2 was highly up-regulated in patients and mice with aGVHD, likely as part of the inflammatory response. In recipient mice lacking LCN2 in the epithelial compartment, more severe inflammation of the gut and liver was observed. These findings support the concept that cells in the recipient compartment contribute to the protective effect of LCN2.

Different studies have reported on the microbiome composition in the gastrointestinal tract of wild-type versus LCN2^{-/-} mice. One study found no difference between wild-type and LCN2^{-/-} mice

with respect to the microbiome (30). Conversely, another study showed that LCN2^{-/-} mice exhibited dysbiosis and persistent colonization with segmented filamentous bacteria (55). The mice used in our study were reported to have no microbiome alteration and no preexisting intestinal inflammation (30), which we confirmed.

The intestinal barrier of patients after allo-HCT is frequently impaired because of the conditioning regimen, antibiotic-induced dysbiosis, and aGVHD-triggered intestinal inflammation (56). Conversely, preserved microbiome diversity at the time of hematopoietic cell engraftment is associated with lower mortality (56). In this

regard, LCN2 might not only exert antibacterial properties but also further promote immune cell migration and mucosal regeneration (25), thereby constituting a vital barrier-maintaining molecule.

We found microbiota changes at the phylum level in the wild-type BM donor versus LCN2^{-/-} BM donor groups, specifically a decreased abundance of *Firmicutes* in mice receiving the LCN2^{-/-} BM. Loss of *Firmicutes* was previously reported to be associated with more severe aGVHD (35). In addition, we observed that the class *Clostridia*, particularly the family *Lachnospiraceae*, was reduced in both the LCN2^{-/-} BM donor group and in LCN2^{-/-}-recipient mice, which is in line with previous studies (36, 37). The interaction of the microbiome with the host immune system has been shown to affect the outcome of patients undergoing allo-HCT. The commensal microbiome of patients after allo-HCT is frequently characterized by a reduced diversity and alterations of the microbial composition, which has been linked to aggravation of intestinal aGVHD (57–59). In contrast to the effects of LCN2 loss, treatment with LCN2 had only minor effects on the intestinal microbiome composition of allo-HCT recipients, and antibiotics treatment did not alter treatment efficacy. Therefore, we conclude that treatment with LCN2 on day 7 does not act mainly through microbiota modification but rather directly on immune cells.

Others showed that LCN2 reduced intestinal inflammation in mice with IBD by enhancing bacterial clearance due to increased phagocytic activity of macrophages (31). In patients with IBD, LCN2 was identified as an indicator of disease activity (28, 29). Likewise, we found that intestinal LCN2 mRNA and LCN2 protein were increased in patients with aGVHD grades 2 to 4 compared with allo-HCT patients without aGVHD. Consistent with our findings, a previous study reported an increase of LCN2 in the serum of aGVHD patients, with higher concentration correlating with increased aGVHD severity and mortality (60). This is in line with other studies showing that blood and tissue expression of protective molecules including GLP-2 and regenerating islet-derived protein 3 alpha (REG3α) correlate with aGVHD severity and resistance to therapy in humans (12, 61). Moreover, immunosuppressive enzymes including IDO increased in the intestinal tissue of patients with aGVHD (62), which can be explained by tissue damage, leading to the release of immunomodulatory molecules as an attempt of the immune system to counterbalance inflammation.

LCN2 has been studied in other pathologies, including malignancies, metabolic disorders, kidney injury, and inflammatory diseases affecting the gastrointestinal tract, lungs, or skin (63, 64). LCN2 had disease-promoting or disease-inhibitory effects depending on the cellular and disease-specific context. For instance, in psoriasis, LCN2 was increased in the serum and the skin (65). A preclinical study using a mouse model of psoriasis showed that wound healing improved with anti-LCN2 antibody treatment (66). Conversely, we observed that LCN2 is protective against skin aGVHD. The difference may be explained by the dissimilar pathomechanisms causing the autoreactive disease psoriasis versus the alloreactive disease aGVHD (2, 67, 68). Functionally, LCN2 induced increased IGF-1R phosphorylation in LCN2-exposed BMDMs, consistent with a previous study showing that the activation of IGF-1R leads to the production of IL-10 in progressive experimental visceral leishmaniasis (69). IGF and IGF-BP expression was found to be lower in patients undergoing allo-HCT compared with controls (70). This suggests a lack of IGF-1R signaling in allo-HCT patients, which could be counterbalanced by LCN2, leading to restored IGF-1R signaling.

Our study has some limitations. Although our finding that LCN2 reduces aGVHD in an IL-10–dependent manner was functionally proven in mice, it is now unclear whether this finding can be directly translated into the clinic for patients with aGVHD. Furthermore, the toxicity profile of LCN2 in patients needs to be studied before a larger clinical trial can be launched. In summary, we identified LCN2 as a beneficial peptide with distinct and context-dependent properties during aGVHD and showed that LCN2⁺ neutrophils can reduce aGVHD severity. Mechanistically, neutrophil-derived LCN2 caused IGF-1R activation in macrophages and induced *Il-10* expression while reducing MHCII expression. In addition, non-hematopoietic LCN2 protected mice from disease exacerbation. Together, these data suggest that LCN2 treatment might constitute a therapeutic approach for aGVHD.

MATERIALS AND METHODS

Study design

In vivo experiments were carried out in accordance to the Animal Research: Reporting of In Vivo Experiments (ARRIVE) guidelines. To investigate the protective functions of neutrophils in the late stages of aGVHD, we used an established murine aGVHD model and profiled the neutrophil transcriptome. We also generated treatment schemes and genetic and adoptive transfer models to decipher the mechanism of LCN2-mediated protection from aGVHD. 16S rRNA gene sequencing was used to determine the effect of LCN2 on the intestinal microbiome. In addition, we used kinase profiling and microarray analyses as independent approaches on which to base in vitro experiments. For all transplantation experiments, we used 6- to 10-week-old age-matched littermates that were randomized into experimental groups before the start of the experiment, without additional measures taken for randomization. The mice were kept under specific pathogen-free conditions in individually ventilated cages at the mouse facilities of Freiburg Medical Center, Innsbruck and the University of California San Diego. All experiments were independent of each other and performed two or three times. For experiments in which we determined the expression of certain molecules by flow cytometry and experiments where the aGVHD severity grade was determined by histology, relevant effect sizes were calculated on the basis of previous experiments (21, 37), resulting in a statistical power of 80% and a relative effect size of 1.2. In these experiments, animals were euthanized at predetermined time points. The endpoints for those outcome measures were reached at day 7, 14, 21, or 27 after allo-HCT in our aGVHD models, respectively; on day 19 in our primary AML model; or on day 8 after induction of colitis. For this study, we had previous knowledge on the disease model and its outcome measures when using vehicle treatment but no data for the experimental intervention (e.g., LCN2 treatment). Thus, when designing the experiments, we built on our experience from previous projects to estimate appropriate effect sizes for biological relevance in the parameters we measured, setting the effect size to 1.2. For survival experiments, an effect size with a hazard ratio higher 3 was considered. In these experiments, the primary outcome was defined by specific elimination criteria or when mice were found dead within the first 80 days after allo-HCT. We did not exclude outliers, unless justified by technical difficulties during data collection. The survival experiments were performed in a non-blinded fashion. The histology scoring experiments were performed in a blinded fashion. To study LCN2 expression in aGVHD patients, we analyzed sample cohorts

from University Medical Center Freiburg, University Regensburg, University Minnesota, and Medical University Innsbruck. Data shown in Figs. 1K and 5 (A to J) were acquired in San Diego, USA; all other mouse experiments were performed in Freiburg, Germany. Relevant effect sizes were calculated on the basis of previous experiments, resulting in a statistical power of 80% and a relative effect size of 1.2. Death is the outcome, 100% (100 per 100) in the vehicle treatment group were expected to die, and we expected that 40% (40 per 100) in the comparison group would die. Therefore, the risk ratio (a measure of the relative effect) is $40/100 = 0.40$.

Statistical analyses

All individual-level data for experiments where $n < 20$ are presented in data files S1 and S2. For the sample size in the murine aGVHD survival experiments, a power analysis was performed, assuming confirmatory study conditions (study design to confirm a hypothesis). Differences in animal survival (Kaplan-Meier survival curves) were analyzed by Mantel-Cox test. Statistical analyses were performed using GraphPad Prism v10 software. Where data did not conform to normal distribution by Shapiro-Wilk test, Mann-Whitney U test or Kruskal-Wallis test with Dunn's multiple comparisons test was used. For normally distributed data, two-tailed unpaired Student's t test or one-way analysis of variance (ANOVA) with Holm-Šidák's multiple comparisons test was used. Paired analyses were performed where appropriate. Data are presented as means $\pm$ SEM. A P value of ≤ 0.05 was considered significant.

Supplementary Materials

This PDF file includes:

Materials and Methods
Figs. S1 to S10
Tables S1 to S11
References (71–88)

Other Supplementary Material for this manuscript includes the following:

Data files S1 and S2
MDAR Reproducibility Checklist

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Lipocalin-2 expression identifies an intestinal regulatory neutrophil population during acute graft-versus-host disease

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Editor's summary

An approach to treating aGVHD. Acute graft-versus-host disease (aGVHD) is a major complication after allogeneic hematopoietic cell transplant (allo-HCT) for which there are limited treatment options. As aGVHD often presents with gastrointestinal (GI) tract inflammation, studying drivers of disease in the GI tract may identify new therapeutic avenues. Here, Czech *et al.* identified a population of lipocalin-2 (LCN2)–expressing neutrophils that were increased in the GI tract of mice with aGVHD and exhibited a tolerogenic effect by promoting IL-10 production and decreasing MHCII expression on macrophages; similar observations were made across multiple cohorts of patients who received allo-HCT. Transfer of LCN2-overexpressing neutrophils or administration of recombinant LCN2 reduced aGVHD severity across multiple murine allo-HCT models, supporting further development of LCN2 as a therapeutic for aGVHD. —Courtney Malo

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